MAX PLANCK INSTITUTE
FOR PSYCHOLINGUISTICS



REPLICATION IN GLOBAL COLLABORATIONS

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Meng-Yun Wang

Low replication rate sends shocking waves

RESEARCH

RESEARCH ARTICLE

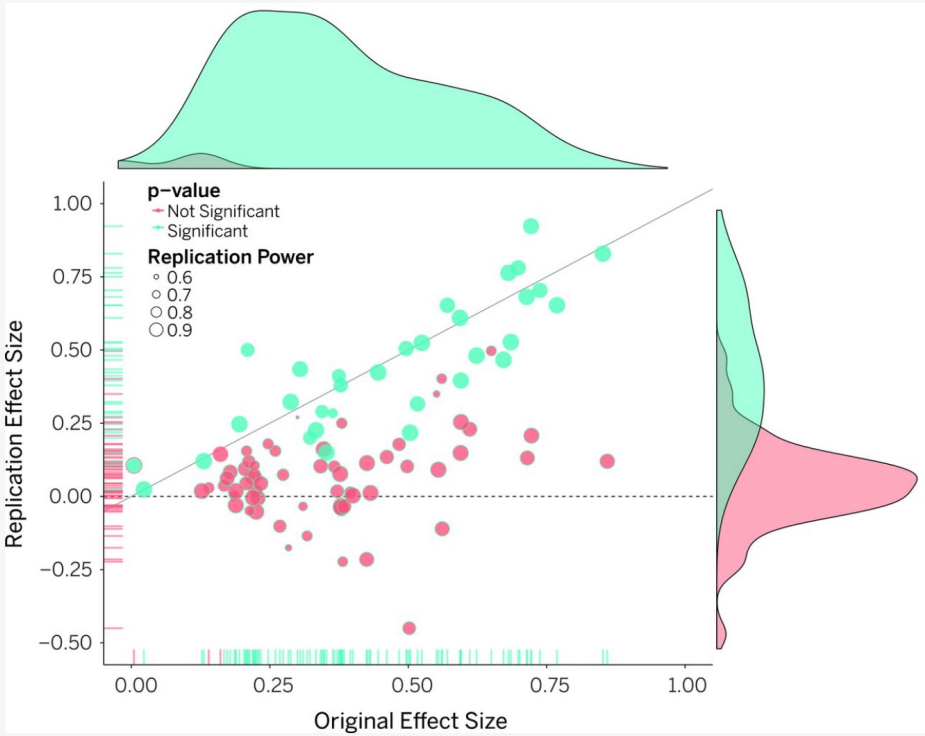
PSYCHOLOGY

Estimating the reproducibility of psychological science

Open Science Collaboration*†

Reproducibility is a defining feature of science, but the extent to which it characterizes current research is unknown. We conducted replications of 100 experimental and correlational studies published in three psychology journals using high-powered designs and original materials when available. Replication effects were half the magnitude of original effects, representing a substantial decline. Ninety-seven percent of original studies had statistically significant results. Thirty-six percent of replications had statistically significant results; 47% of original effect sizes were in the 95% confidence interval of the replication effect size; 39% of effects were subjectively rated to have replicated the original result; and if no bias in original results is assumed, combining original and replication results left 68% with statistically significant effects. Correlational tests suggest that replication success was better predicted by the strength of original evidence than by characteristics of the original and replication teams.

Only 1/3 to 1/2 of the original findings were also observed in the replication study.



Original study effect size versus replication effect size (correlation coefficients). Diagonal line represents replication effect size equal to original effect size. Dotted line represents replication effect size of 0. Points below the dotted line were effects in the opposite direction of the original. Density plots are separated by significant (blue) and nonsignificant (red) effects.

Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, *349*(6251), aac4716.

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Pipeline choices alter fMRI findings

Article

Variability in the analysis of a single neuroimaging dataset by many teams

https://doi.org/10.1038/s41586-020-2314-9

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Check for updates

A list of authors and affiliations appears in the online version of the paper.

Data analysis workflows in many scientific domains have become increasingly complex and flexible. Here we assess the effect of this flexibility on the results of functional magnetic resonance imaging by asking 70 independent teams to analyse the same dataset, testing the same 9 ex-ante hypotheses[1]. The flexibility of analytical approaches is exemplified by the fact that no two teams chose identical workflows to analyse the data. This flexibility resulted in sizeable variation in the results of hypothesis tests, even for teams whose statistical maps were highly correlated at intermediate stages of the analysis pipeline. Variation in reported results was related to several aspects of analysis methodology. Notably, a meta-analytical approach that aggregated information across teams yielded a significant consensus in activated regions. Furthermore, prediction markets of researchers in the field revealed an overestimation of the likelihood of significant findings, even by researchers with direct knowledge of the dataset[2–5]. Our findings show that analytical flexibility can have substantial effects on scientific conclusions, and identify factors that may be related to variability in the analysis of functional magnetic resonance imaging. The results emphasize the importance of validating and sharing complex analysis workflows, and demonstrate the need for performing and reporting multiple analyses of the same data. Potential approaches that could be used to mitigate issues related to analytical variability are discussed.

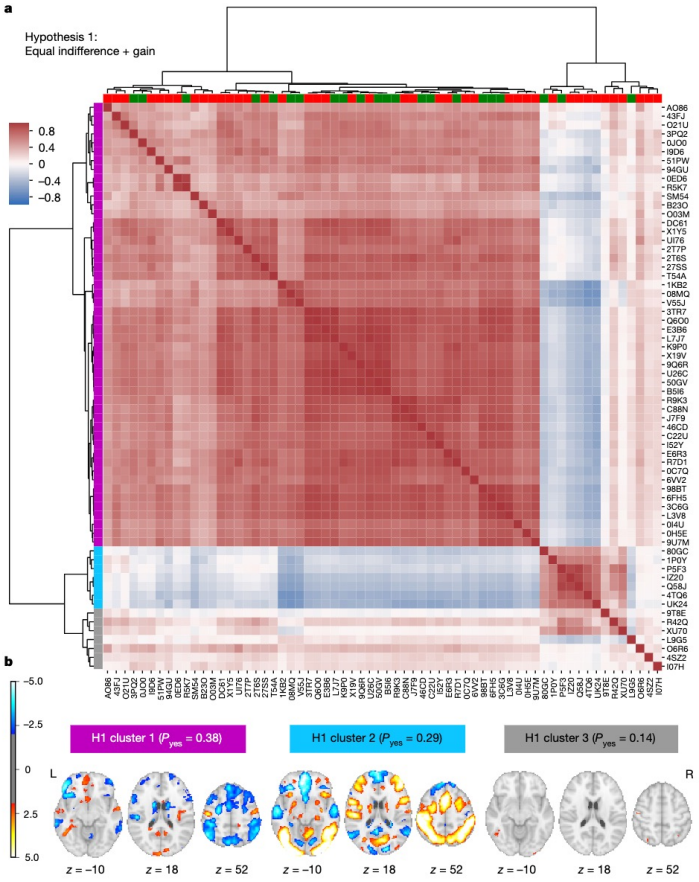


Fig. 2 | Analytical variability in whole-brain statistical results for hypothesis 1 (and hypothesis 3). a, Spearman correlation values between whole-brain unthresholded statistical maps for each team ($n = 64$) were computed and clustered according to their similarity (using Ward clustering on Euclidean distances). Row colours (left) denote cluster membership (purple, cluster 1; blue, cluster 2; grey, cluster 3); column colours (top) represent hypothesis decisions (green, yes; red, no). Brackets represent clustering. **b**, Average statistical maps (thresholded at uncorrected $z > 2.0$) for each of the three clusters shown on the left in **a**. The probability of reporting a positive hypothesis outcome (P_{yes}) is presented for each cluster. L, left; R, right. Unthresholded maps for hypotheses 1 and 3 are identical (as they both relate to the same contrast and group but different regions), and the colours represent reported results for hypothesis 1. Images can be viewed at https://identifiers.org/neurovault.collection:6048.

70 laboratories that analysed the same neuroimaging data each produced different results.

Botvinik-Nezer, R., Holzmeister, F., Camerer, C. F., Dreber, A., Huber, J., Johannesson, M., ... & Rieck, J. R. (2020). Variability in the analysis of a single neuroimaging dataset by many teams. *Nature*, *582*(7810), 84-88.

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EEGManyPipelines project

396 scientists (168 teams) analysed the same EEG data each produced different results.

ERP variability across teams was equivalent to variability across participants, suggesting that **who analyses the data** is as important as who provides it.

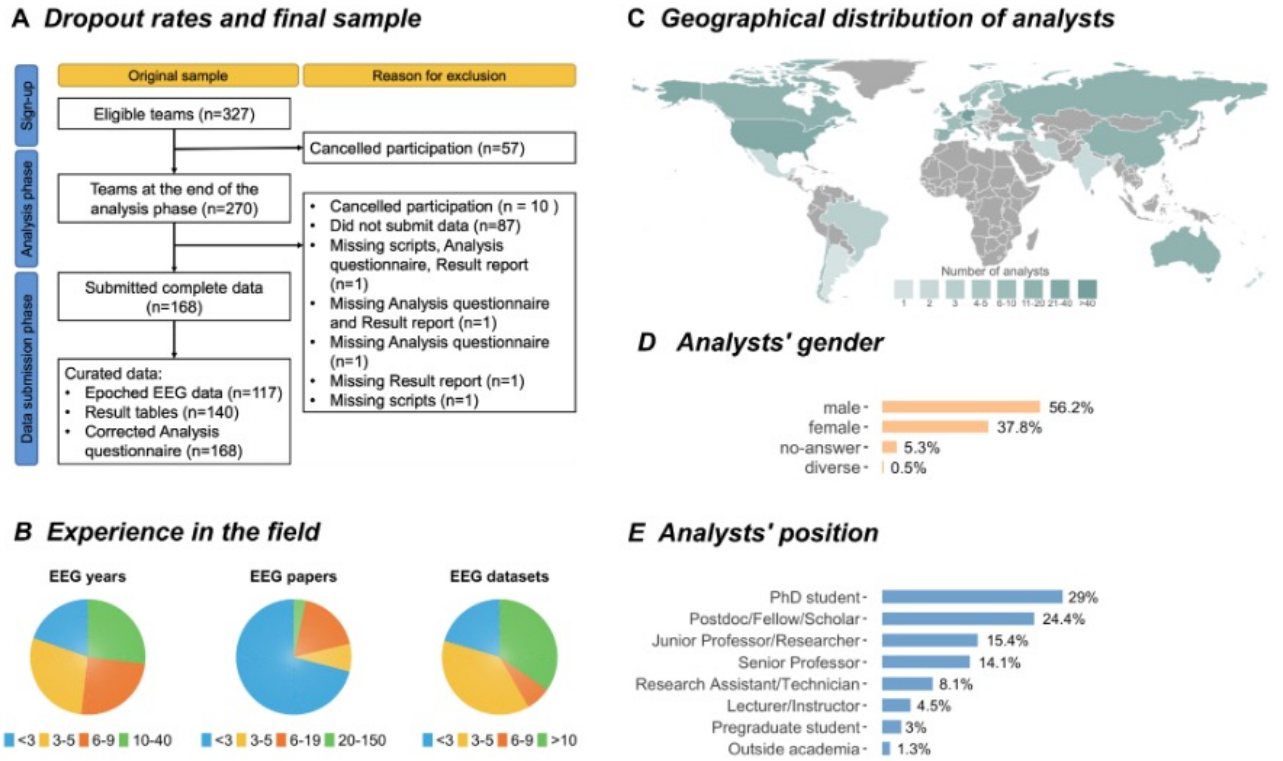


Figure 1. Inclusion flowchart and demographic information of participating analysts (n=168). A. Flow diagram indicating team numbers at different stages and reasons for dropout. **B.** Expertise in performing EEG, the number of published EEG papers, and the number of collected EEG datasets. **C.-E.** Charts summarize demographic data of the participating analysts: geographical distribution of analysts (**C**), gender (**D**), and self-reported job position grouped into categories (**E**). The figure in panel C was adapted from Trübutschek et al. (2024).

Cesnaite, E., Pascarella, A., Vinding, M., Algermissen, J., Fischer, N., Puoliväli, T., ... & Trübutschek, D. (2025). The EEGManyPipelines Dataset: Metascientific Data on 168 Independent Analyses of a Single EEG Dataset.

committee@eegmanypipelines.org Inbox - UiB-Wang 2 September 2021 at 14:37
Registration to the EEGManyPipelines project
To: Mengyun Wang

Dear Analyst,

Thank you for registering to the EEGManyPipelines project!

Your team identifier is 4dc16bcffc6daf9b.

The EEG data becomes available for download after the registration period (DL: October 3rd 2021) has passed.

For questions or comments, please write to committee@eegmanypipelines.org

Best regards,
The EEGManyPipelines steering committee

committee@eegmanypipelines.org Inbox - UiB-Wang 8 October 2021 at 14:09
Instructions for downloading the EEGManyPipelines data
To: committee@eegmanypipelines.org

Dear Analyst,

We are excited to have you onboard and look forward to working with you. You successfully registered and are now part of a group of more than 300 teams, including more than 600 researchers from around the world.

This email contains the necessary information to get your data analysis started. Please follow these four steps:

committee@eegmanypipelines.org Inbox - UiB-Wang 13 May 2022 at 22:09
EEGManyPipelines: few days left to submit your data
To: committee@eegmanypipelines.org

Dear Analyst,

This is a kind reminder that there are only a few days left to upload your data before the submission deadline on May 15th (if you have already submitted your data and results, this email does not concern you). Please contact us by email committee@eegmanypipelines.org in case you are experiencing any problems uploading your data, or if you have not received the instructions for submitting your data and answering the structured questionnaires.

If you don't know how to proceed with your data upload or questionnaires, please have a look at our FAQ section (https://www.eegmanypipelines.org/faq.php). We are continuously updating it to cover the most frequently asked questions.

After the submission deadline, we will check all data for completeness and come back to you with either a confirmation email or a request to complete your submission if something is missing.

Thank you for participating in the EEGManyPipelines project!

Best regards,
The EEGManyPipelines Steering committee

Steering Committee Inbox - UiB-Wang 20 November 2024 at 18:10
EEGManyPipelines: first results and an invitation to comment
To: undisclosed-recipients:;, Bcc: Mengyun Wang Details

Dear Analysts,

Greetings from the EEGManyPipelines team!

In the last two years, we have systematically assessed the reported pipelines, checked all the submitted code, and brought the submitted preprocessed data into a common format. This has been a lot of work, but it has finally borne fruit! We are now in the position where we can start to map the effect of analysis choices onto the results of each analysis team (comprising both self-reported results and pre-processed EEG data).

At this critical stage in our analysis, we're excited to share our initial progress and analytical approaches to this rich EEG dataset with you as a co-author. We completed a detailed analysis of the first hypothesis of the eight hypotheses, summarized in a 14-page document. At the moment, we are further extending this work to hypotheses 2a and 3a. Please note that this is not a draft of the paper, but a document summarizing preliminary findings.

EEGManyPipelines Inbox - MPI 1 July 2026 at 18:08
Update on the EEGManyPipelines manuscripts
To: undisclosed-recipients:;

Dear all,

It's been a busy couple of months for the project after both EEGManyPipelines manuscripts have been submitted to Nature Human Behaviour. We would like to share with you an update on the current status of the papers.

-The main EEGManyPipelines manuscript

We are happy to share that the main EEGManyPipelines manuscript went under review, and we received rather positive feedback. We are currently working on revising the manuscript and addressing the reviewers' comments. Once it's ready, we'll share both the reply to reviewers and the revised manuscript with all co-authors.

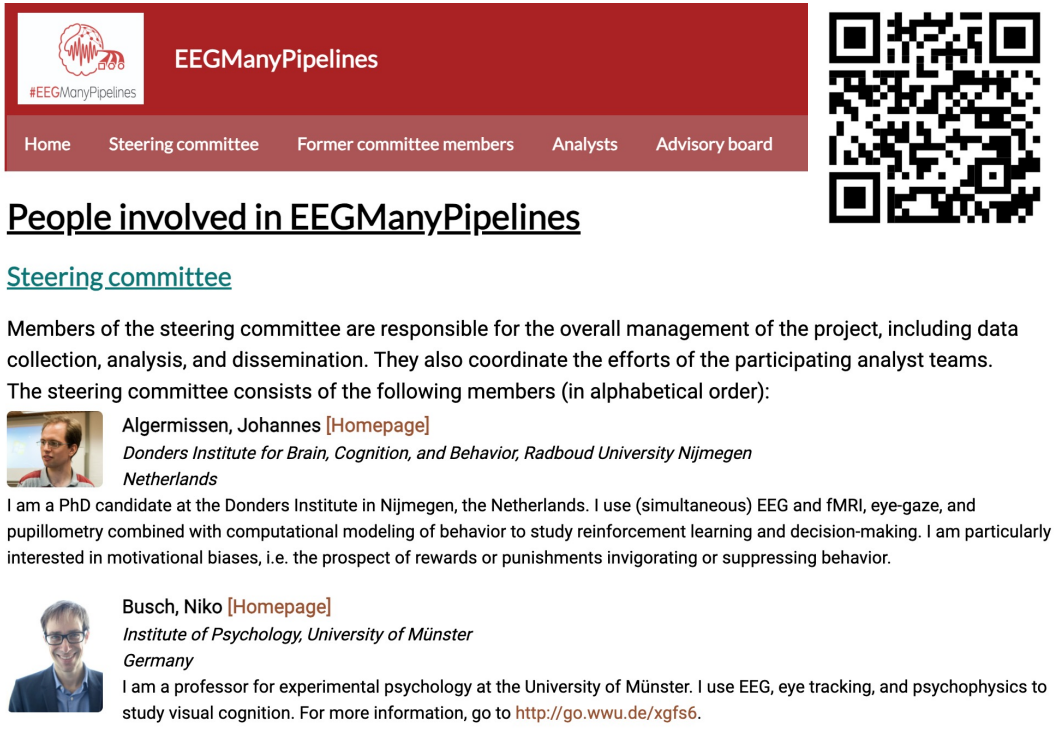
-The Data Descriptor paper

We also received a suggestion from the editor at Nature Human Behavior to resubmit the Data Descriptor paper to Communications Psychology as a Resource article. To do so, we had to modify its structure and add 'Example uses' and 'Discussion' sections. The updated version of the Data descriptor paper can be found here: https://docs.google.com/document/d/1ltofF1mK6hVA4h4-SSoKNYQekWqNpG_z4JxtkNgqruA/edit?tab=t.0

What I have learned

- What I enjoyed most was how clearly the project was organised.
- The instructions and outlines were clear and well thought out.
- I also really appreciated having someone keeping things moving.
- With so many people involved, it helps to have a team setting the agenda, sending reminders, and following up.

- The downside was the timeline. It did drag on a bit, and sometimes I found it hard to keep up with all the results and updates.



EEGManyPipelines

#EEGManyPipelines

Home | Steering committee | Former committee members | Analysts | Advisory board

People involved in EEGManyPipelines

Steering committee

Members of the steering committee are responsible for the overall management of the project, including data collection, analysis, and dissemination. They also coordinate the efforts of the participating analyst teams. The steering committee consists of the following members (in alphabetical order):

Algermissen, Johannes [Homepage]
Donders Institute for Brain, Cognition, and Behavior, Radboud University Nijmegen
Netherlands
I am a PhD candidate at the Donders Institute in Nijmegen, the Netherlands. I use (simultaneous) EEG and fMRI, eye-gaze, and pupillometry combined with computational modeling of behavior to study reinforcement learning and decision-making. I am particularly interested in motivational biases, i.e. the prospect of rewards or punishments invigorating or suppressing behavior.

Busch, Niko [Homepage]
Institute of Psychology, University of Münster
Germany
I am a professor for experimental psychology at the University of Münster. I use EEG, eye tracking, and psychophysics to study visual cognition. For more information, go to http://go.wwu.de/xgfs6.

https://eegmanypipelines.github.io/index.html

Thank you
for your attention

Meng-Yun Wang
mengyun.wang@mpi.nl

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